

Fabrication of zwitterionic polymer-functionalized MXene nanosheets for anti-bacterial and anti-biofouling applications

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ABSTRACT

Marine biological pollution is a worldwide problem that poses a significant challenge in marine engineering. MXene nanosheets and zwitterionic polymers are promising materials for marine anti-biofouling and have attracted much attention. Herein, we self-assembled the polydopamine layer on the surface of MXene nanosheets (MXene-PDA), and then introduced the zwitterionic polymer (PEIS) by Michael addition reaction onto the PDA layer to obtain the zwitterionic polymer-functionalized MXene nanosheets (MXene-PEIS). The as-prepared MXene-PEIS exhibits excellent antifouling performance through the cooperative antifouling effect between the MXene nanosheets and the zwitterionic polymer PEIS (the antibacterial rate improved by 88 % and there was a 98 % decrease in microalgae adhesion density). Moreover, the nanocomposite antifouling coatings (MXene-PEIS/SP) were obtained by blending the as-prepared MXene-PEIS nanosheets with self-polishing coatings. The results show that MXene-PEIS/SP coatings have good antifouling performance due to the synergy between MXene, PEIS, and self-polishing coatings.

1. Introduction

Marine biological pollution is a complex issue that not only causes significant economic losses and security issues, but also has a negative impact on marine ecosystems. Marine biological pollution refers to the undesirable adhesion of biological and abiotic dissolved compounds, microorganisms, plants, and animals on ship hulls and equipment immersed in seawater [1], which tends to accelerate the deterioration of surface corrosion, cause structural and functional defects, increase navigational resistance, lead to higher fuel consumption, increased maintenance costs, and so on [2]. Furthermore, marine fouling species attached to the surface of the ship may be carried to another area along with the movement of the ship, resulting in the invasion of alien species and the destruction of the local ecological balance, thus causing significant loss to fisheries [3]. As a result, the advancement of marine anti-fouling technology is critical for economic development, marine resource exploitation and utilization, and environmental protection.

Antifouling coatings are widely used in marine antifouling because of their ease of use, low cost, and high efficiency. There are a variety of natural antifouling surfaces found in animals and plants. Based on the bionics, researchers have designed diversified antifouling coatings, such

as antifouling agent-releasing coating [4], dynamic surface antifouling coating [5], and fouling-resistant coating [6], through physical or chemical methods to prevent the surface from being attached and colonized by marine fouling organisms, or by making it easy to separate them from the surface, to achieve antifouling [3]. Antifouling coatings with the Cu₂O antifouling agent are widely applied in marine equipment at present. The copper ions produced by Cu₂O in seawater can deactivate the main enzymes that marine organisms depend on for survival, leading to changes in biological tissues, followed by death, thus presenting antifouling performance. However, the long-term release of copper in the marine environment is potentially harmful to the marine ecological balance [7]. As a results, it is critical to develop antifouling coatings that are more efficient, long-lasting, and safe for the environment.

The self-polishing antifouling coating hydrolyzes in seawater, releasing the antifouling agent steadily and continuously [8]. The synergy between the antifouling agent and the dynamic renewal of the coating surface contributes to excellent and long-lasting antifouling performance. At the same time, the self-polishing coating can be easily incorporated in the production processes. Therefore, the improvement and extension of antifouling coating performance can be realized in a viable manner by developing antifouling agents with good performance

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that have self-polishing coatings.

MXene is a novel two-dimensional carbon/nitride nanolayered material; its formula can be expressed as $M_{n+1}X_nT_x$, where M represents the transition metal element, X is carbon or nitrogen, n is generally 1, 2, or 3, and T_x refers to surface terminating functional groups (OH, F, O, etc.) produced by chemical etching [9]. MXene not only has the advantages of traditional two-dimensional materials, but also has prominent characteristics such as good hydrophilicity, environmental friendliness [10], and adjustable structure [11], thus presenting great application potential. In recent years, the antibacterial performance of MXene has attracted great attention. Studies have shown that $Ti_3C_2T_x$ can destroy cell membranes and lead to cell injury and death when cells are placed in direct contact with MXene due to its unique physicochemical properties and excellent antibacterial performance [12]. There are a wide variety of biomedical applications of MXenes, including drug delivery, cancer therapy, and biosensing [13]. As an emerging antibacterial nanomaterial of great potential, further development of antifouling applications of MXenes is promising. Wang et al. modified $Ti_3C_2T_x$ with a block copolymer (PMBTMA-co-PSPMA) which was formed by the sequential grafting of anticorrosion agent (PMBTMA) and antifouling agent (PSPMA) through surface-initiated atom transfer radical polymerization [14]. An antifouling and anticorrosive coating was achieved by mixing with waterborne epoxy resin.

Zwitterionic polymers are a new type of antifouling material that exhibit significant hydration ability and effectively inhibit the adhesion of proteins and bacteria by forming a hydration layer [15]—this principle is widely employed in the design of antifouling surfaces. Mussel-inspired catechol chemistry provides many simple and attractive ways for surface modification of materials, especially for the anchoring of zwitterionic polymers [16]. Catechol and its derivatives have been successfully involved in synthesizing various polymers as the “graft to” anchor groups [17]; they also provide an anchoring platform for the polymer by modifying the surface of the material, so as to achieve covalent connection of the zwitterionic polymer on the material surface. Kallem et al. prepared zwitterion-functionalized MXene nanosheets with PDA and then grafting taurine, the zwitterion-grafted MXene nanocomposite membranes displayed the improved water permeability and good antifouling properties [18]. Compared to zwitterion with low molecular weight such as taurine, zwitterionic polymers are more conducive to the formation of a denser modification layer [19]. He et al. introduced zwitterionic antifouling polymer (PEIS) into Gallium-based liquid metal nanodroplets modified with PDA (PEIS-GLM) through surface modification, and proved that PEIS-GLM has significant antibacterial and antifouling properties among polydimethylsiloxane (PDMS) coatings [20]. Therefore, it is believed that the grafting of PEIS can improve both antibacterial and antifouling properties due to the introduction of quaternary ammonium groups, and zwitterionic groups [21–23].

Inspired by various methods of functioning of the antifouling surface, we prepared zwitterionic polymer (PEIS)-functionalized MXene nanosheets through a two-step modification method. We self-assembled the polydopamine layer on the surface of MXene nanosheets (MXene-PDA), after which we introduced the quaternized polyethylenimine derivative (PEIS) through Michael addition onto the PDA layer to obtain the zwitterionic polymer-functionalized MXene nanosheets (MXene-PEIS). Subsequently, the experimental results revealed that the as-prepared MXene-PEIS nanosheets exhibited significant antibacterial and antibiofouling properties. The nanocomposite antifouling coatings were obtained by blending the as-prepared MXene-PEIS nanosheets with the self-polishing coatings (MXene-PEIS/SP). The experimental results demonstrated that MXene-PEIS/SP coatings also presented good antifouling performance. In addition, the MXene-PEIS nanosheets played a positive role in prolonging the service life of self-polishing coatings.

2. Experimental section

2.1. Materials

Sodium fluoride (NaF), Hydrochloric acid (HCl), and Ammonium fluoride (NH_4F) were obtained from Sinopharm Chemical Reagent Co., Ltd. Ti_3AlC_2 (98 %) was obtained from Ningbo Beijiaer New Material Co. Ltd. Dopamine hydrochloride (98 %), 1,3-propanesultone (99 %), polyethylenimine (PEI, M.W.10000), and tris(hydroxymethyl)aminomethane (Tris, 99 %) were purchased from Aladdin Industrial Corporation. Self-polishing resin was obtained from Luojiang Sanxing Paint Resin Co., Ltd. The deionized water (DI water) was used as received. Quaternized polyethylenimine derivative (PEIS) was synthesized by the process given in a previous study [24].

2.2. Preparation of the $Ti_3C_2T_x$ MXene nanosheets

$Ti_3C_2T_x$ MXene was synthesized referring to the procedure given in a preceding study [25]. Firstly, 1.0 g NaF and 1.0 g NH_4F were stirred in 20 mL HCl (6 M) until completely dissolved. Next, 2.0 g Ti_3AlC_2 powder was gradually added to be etched at 65 °C and a 500 rpm stirring speed for 100 h. After the etching was complete, the as-prepared MXene suspension liquid was repeatedly centrifuged and diluted with DI water until the diluted solution approached a neutral pH not less than 6. Subsequently, MXene was vacuum freeze-dried for two days and the $Ti_3C_2T_x$ MXene powder was finally collected for the experiment.

2.3. Synthesis of zwitterionic polymer-functionalized MXene nanosheets

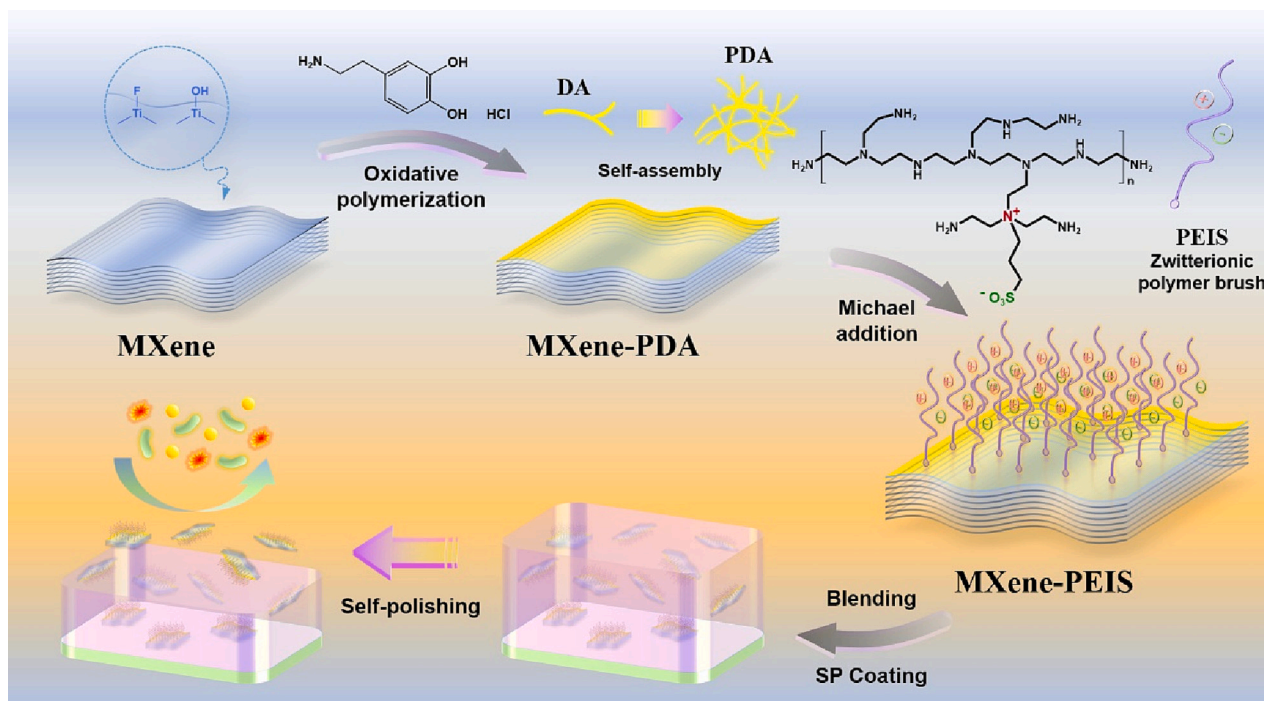
A two-step modification method was utilized to synthesize the PEIS-functionalized MXene nanosheets, as the Scheme 1 shows. Initially, 0.2 g MXene powder was dispersed into 50 mL DI water and sonicated for better dispersion. Next, the pH of the as-prepared suspension liquid was raised to 8.5 by mixing with 60.5 mg tris(hydroxymethyl)aminomethane in a 100 mL round-bottomed flask. After the addition of 15 mg dopamine hydrochloride, the reaction was carried out under continuous stirring at room temperature overnight to obtain MXene-PDA. Subsequently, PEIS polymer brushes were grafted to MXene-PDA via Michael addition. Then, 150 mg MXene-PDA nanosheets were dispersed in 50 mL Tris buffer (10 mmolL⁻¹, pH 8.5), followed by the addition of 50 mg PEIS. After a 24-h stirring at room temperature, the as-prepared MXene-PEIS nanosheets were prepared into powder through vacuum drying.

2.4. Preparation of self-polishing nanocomposite coating

The nanocomposite coating was prepared as follows: 20 mg MXene-PEIS nanosheets were added to 2 g self-polishing polymer(silyl acrylate polymer) with a quick and even stirring. The as-prepared mixture (1 wt %) was uniformly applied on a clean glass slide and subsequently dried for 48 h at 25 °C. The final self-polishing nanocomposite coatings were named MXene-PEIS/SP.

2.5. Characterization

X-ray photoelectron spectroscopy (XPS) spectra were carried out on a PHI VersaProbe III. Fourier transform infrared (FT-IR) spectra were used to investigate the chemical compositions of the samples. Zeta potential of the samples was analyzed by a Malvern Zetasizer Nano zeta potential analyzer. Thermal stability was measured with a Netzsch STA 449F3 thermogravimetric analyzer (TGA) at the temperature ranging from 80 °C to 700 °C under a N_2 atmosphere. The micromorphology of several MXene nanosheets were observed by the FEI Talos F200X transmission electron microscope (TEM), while the corresponding energy dispersive spectrometer (EDS) was used to measure element mappings. The fluorescence micrographs of microalgae adhered on the coating surface were



Scheme 1. Preparation process of MXene-PEIS nanosheets and MXene-PEIS/SP antifouling coating.

obtained by means of an Olympus BX51 microscope. Scanning electron microscope (SEM) images of MXene nanosheets and the hydrolysis of coatings were acquired respectively by using a Tescan Clara GMH scanning electron microscope.

The antifouling property of MXene nanosheets was verified by the anti-bacterial activity and anti-biofouling activity tests [26], which were performed referencing a previous study [20].

The anti-bacterial activity test was as follows: the bacterial

(*Staphylococcus aureus* and *Escherichia coli*) solution was prepared to a concentration of 1×10^5 CFU/mL with the phosphate-buffered saline (PBS) solution for further use. Then, the as-prepared MXene nanosheets were mixed into the bacterial solution with a 0.5 wt% additive amount and cultured at 37 °C for 5 h. A sample of the bacterial solution without MXene nanosheets was used as the “control”. Subsequently, 15 μ L bacteria suspensions was seeded on LB agar plates evenly and then kept at 37 °C overnight in a constant temperature incubator for the later

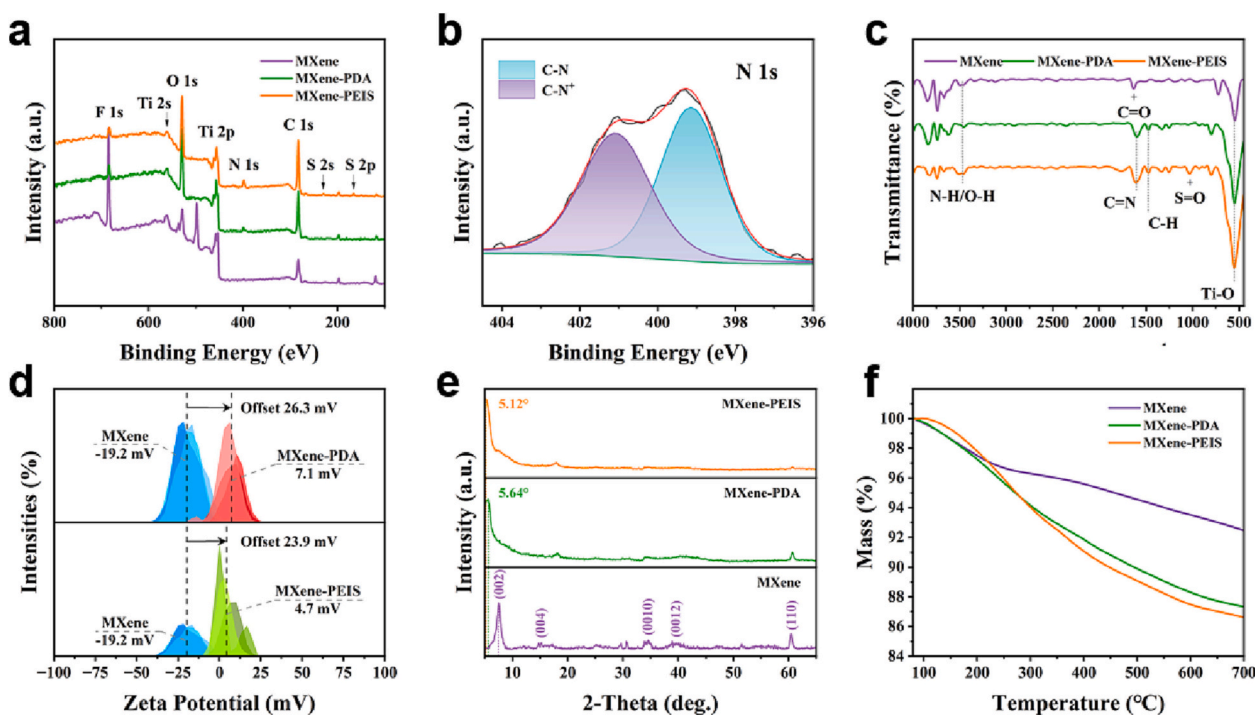


Fig. 1. (a) XPS survey spectra of MXene, MXene-PDA and MXene-PEIS; (b) Curve-fitted XPS spectra of N1s of MXene-PEIS; (c) FT-IR spectra; (d) Zeta potential; (e) XRD patterns; and (f) TG curves of MXene, MXene-PDA, and MXene-PEIS.

observation of the bacteria.

The following is the process of anti-biofouling activity test: the microalgae (*Dunaliella* or *Porphyridium*) suspension was cultured until the concentration reached a density of 10^7 CFU/mL. Next, MXene nanosheets were mixed into the microalgae suspension with a 0.5 wt% additive amount and cultivated at 25 °C overnight. Then, the clean blank PDMS film was cut into a 1 cm × 1 cm piece to be put in the as-prepared mixture and cultivated for 24 h, and the PDMS film of the control group was immersed in the microalgae suspension only. After simple washing to remove the unsettled microalgae, the adhesion condition of microalgae on the PDMS film was evaluated from 10 visual fields which were randomly selected on each PDMS film.

3. Results and discussion

XPS, FT-IR and zeta potential measurement were done to verify the successful grafting of PEIS on the MXene nanosheets. Fig. 1(a) demonstrates the XPS survey spectra of MXene, MXene-PDA, and MXene-PEIS. In the survey spectra of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene, the elemental peaks mainly consist of Ti, C, O, and F [25]. As shown in Fig. 1(a) and Fig. 1(b), the appearance of the S signal and the peak fitting of N1s confirmed the successful grafting of PEIS. The detection of the S2p peak and S2s peak is attributed to the sulfonate ions in PEIS. Meanwhile, the N1s spectra of the MXene-PEIS was curve-fitted into two peaks at 401.1 eV (C-N^+) and 399.2 eV (C-N), indicating the existence of $\text{N}(\text{CH}_3)_4^+$ belonging to PEIS [20,24]. Furthermore, the C1s of MXene, MXene-PDA, and MXene-PEIS are shown in Fig. S8. The C-Ti peak of MXene-PDA is observed a slight shift (0.2 eV) compared to that of MXene, which is possibly derived by the chemical bonding between catecholic group of PDA and the Ti [27]. The FT-IR spectra of MXene, MXene-PDA, and MXene-PEIS are presented in Fig. 1(c). The peaks of unmodified MXene are mainly located at 3441 cm^{-1} , 1634 cm^{-1} , and 554 cm^{-1} , separately relating to the stretching vibrations of O—H, C=O, and Ti—O [14,28]. After modification with PDA, the new absorption peaks are found at 1602 cm^{-1} and 1481 cm^{-1} , which are believed to be associated with the C=N and C—H of PDA [20,29,30]. Besides, the absorption peak at 3441 cm^{-1} is regarded as the stretching vibrations of N—H/O—H [31]. With the introduction of PEIS, a weak peak at 1038 cm^{-1} is observed corresponding to the stretching vibration of S=O [32–34]. When combined with the XPS results, the successful grafting of PEIS was further demonstrated. Moreover, the zeta potential also served as evidence of successful modification of PEIS, as shown in Fig. 1(d). The surface zeta potential of the MXene is -19.2 mV , while the surface zeta potential of the MXene-PDA transforms to 7.1 mV , which is attributed to the amino groups of PDA [24,35–37]. Moreover, the MXene-PEIS shows a relatively neutral zeta potential. It was reported that electrically neutral surfaces are conducive to maximizing the surface hydration and improving the antifouling ability [38]. In addition, Fig. 1(e) shows the XRD patterns of MXene, MXene-PDA, and MXene-PEIS. MXene exhibits typical diffraction peaks at 7.5° , 15.3° , 34.7° , 39.9° , and 60.5° , which are indexed to the (002), (004), (0010), (0012), and (110) peaks, respectively [12,14]. The (002) peaks of MXene-PDA and MXene-PEIS are observed at lower angles of 5.64° and 5.12° , which indicates the increase in interspace between the MXene nanosheets during surface modification [39]; the expanding of the interspace probably derived from the intercalation of polymers into MXene interlayers [25]. The thermo-gravimetric (TG) curves of MXene, MXene-PDA, and MXene-PEIS were analyzed to evaluate the variation in the organic portion. As shown in Fig. 1(f), the weight loss of MXene is 7.37 % in the range of 80 °C to 700 °C, while the weight loss of MXene-PDA increases to 12.39 %; the extra 5.02 % is attributed to the successful modification of PDA. Moreover, the weight loss of MXene-PEIS is 0.89 % more than MXene-PDA due to the successful grafting of PEIS. Furthermore, there is a significant decrease in static water contact angle after the PEIS modification indicating the improvement of hydrophilicity (Fig. S9).

The morphology of MXene, MXene-PDA, and MXene-PEIS were

observed SEM and TEM, as shown in Fig. S1 and Fig. 2. The SEM image of MXene nanosheets demonstrates a clear accordion-like structure with visibly separated layers in Fig. S1(a), indicating the successful fabrication of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene through the etching of Ti_3AlC_2 [40]. Moreover, Fig. S1(b) and Fig. S1(c) show that the modified MXene nanosheets acquired larger interspaces consistent with XRD results and have less layers for more accessible sharp edges, which are beneficial to anti-bacterial activity [13]. MXene nanosheets were prepared for a structure having few layers only, to obtain an easily observable surface. Fig. 2(a) shows the thin and transparent MXene nanosheets with a sharp boundary and rough surface, while the surface of MXene-PDA appears smoother due to the weak conductivity of PDA as shown in Fig. 2(b). The TEM image of MXene-PEIS in Fig. 2(c) presents the further blurring of the boundary related to the grafting of PEIS [37]. As shown in Fig. 2(d), the polymer layer on the nanosheets is about 2–3 nm. To further confirm the PEIS modification, high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of MXene-PEIS and the corresponding EDS mappings were taken. The uniform distribution of S element seen in Fig. 2(i) and Fig. S1(h) indicates the successful modification of PEIS. Additionally, Fig. 2(f)–(i) and Fig. S1(d)–(h) illustrate the distribution of typical elements that provide additional confirmation of the successful fabrication of MXene-PEIS.

The anti-bacterial and anti-biofouling performance of MXene nanosheets were tested to evaluate the antifouling properties, as seen in Figs. 3 and 4. The anti-bacterial activity test was conducted by using two major classes of bacteria (Gram-negative *E. coli* and Gram-positive *S. aureus*), and the colony numbers directly reflect the anti-bacterial effect. As can be seen in Fig. 3(a) and (e), the bacteria multiplied rapidly to form colonies without the addition of MXene nanosheets. Fig. 3(b) and (f) show smaller numbers of colonies than the control group indicating the significant anti-bacterial performance of MXene, as MXene can destroy cell membrane and cause cell injury and death in case of direct contact [12]. The MXene-PEIS group has the best anti-bacterial effect among the experimental groups as shown in Fig. 3(d) and (h). Notably, the antibacterial rate against *E. coli* and *S. aureus* separately improved to 75 % and 88 % (Fig. S2). On the one hand, $\text{Ti}_3\text{C}_2\text{T}_x$ MXene was reported to show excellent antibacterial effect on both *E. coli* and *S. aureus* due to its unique physicochemical properties described in previous research [12,41,42]. On the other hand, PEIS contains the quaternary ammonium group which is generally believed to cause the disruption of the cell membrane [21,43]. Therefore, MXene and quaternary ammonium group of PEIS exert a synergistic effect on the anti-bacterial performance of MXene-PEIS with a significant increase in performance. The test for anti-biofouling activity was conducted by using two typical microalgae (*Dunaliella* and *Porphyridium*), and the anti-biofouling effect was evaluated by the adhesion condition of microalgae on PDMS films viewed through a fluorescence microscope. Fig. 4 shows the fluorescence microscopy images of attached microalgae, and Fig. S3 demonstrated the corresponding statistical results. The control group adhered densely as shown in Fig. 4(a) and (e), while the microalgae of MXene nanosheets groups showed apparent reduction. As shown in Fig. 4 and Fig. S3, the densities of the microalgae further decreased after PDA modification and further reduced substantially after the PEIS modification, especially with a 98 % decrease of *Porphyridium*. In conclusion, the results of anti-biofouling effect against these two types of microalgae (*Dunaliella* and *Porphyridium*) are similar, namely “MXene-PEIS > MXene-PDA > MXene > Control”. The enhancement to anti-biofouling property is attributed to the synergistic effect of the destruction of cells caused by MXene nanosheets and quaternary ammonium group of PEIS, and the protein adsorption-resisting hydration layer that is formed by hydrogen bonding and the solvation of zwitterionic groups of PEIS [23,44]. It follows that the excellent anti-fouling performance of MXene-PEIS is from the dual action of MXene and PEIS.

The as-prepared MXene nanosheets were incorporated into the self-polishing coating to obtain nanocomposite antifouling coatings.

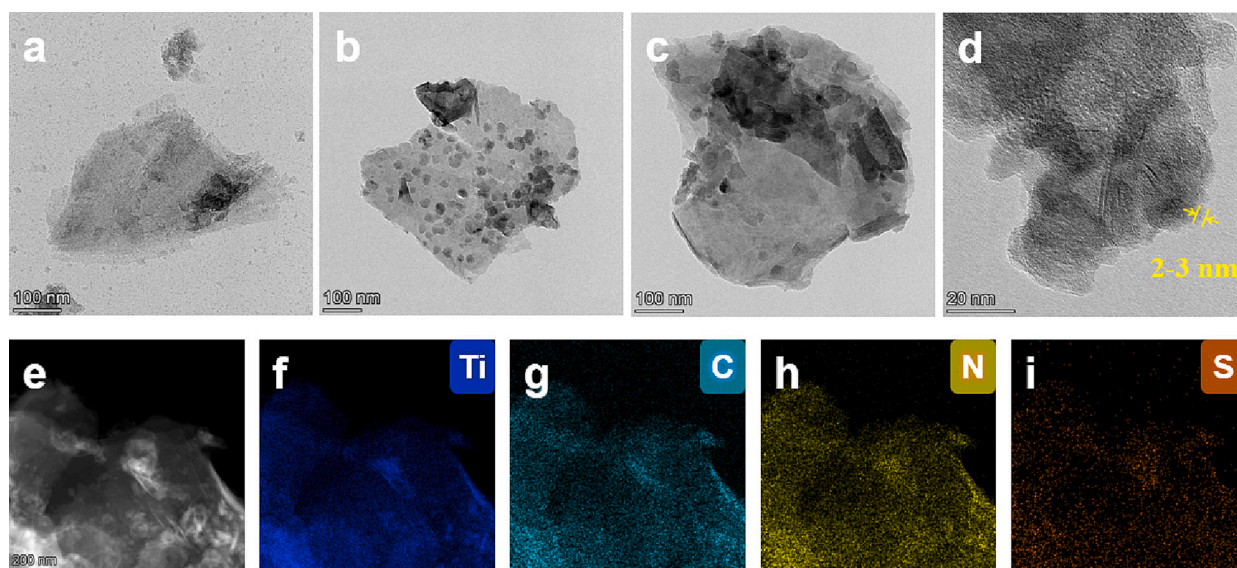


Fig. 2. TEM images of (a) MXene, (b) MXene-PDA, and (c) MXene-PEIS; (d) high-resolution TEM image of MXene-PEIS; (e) HAADF-STEM image of MXene-PEIS; and the corresponding EDS mapping of (f) titanium, (g) carbon, (h) nitrogen and (i) sulfur elements.

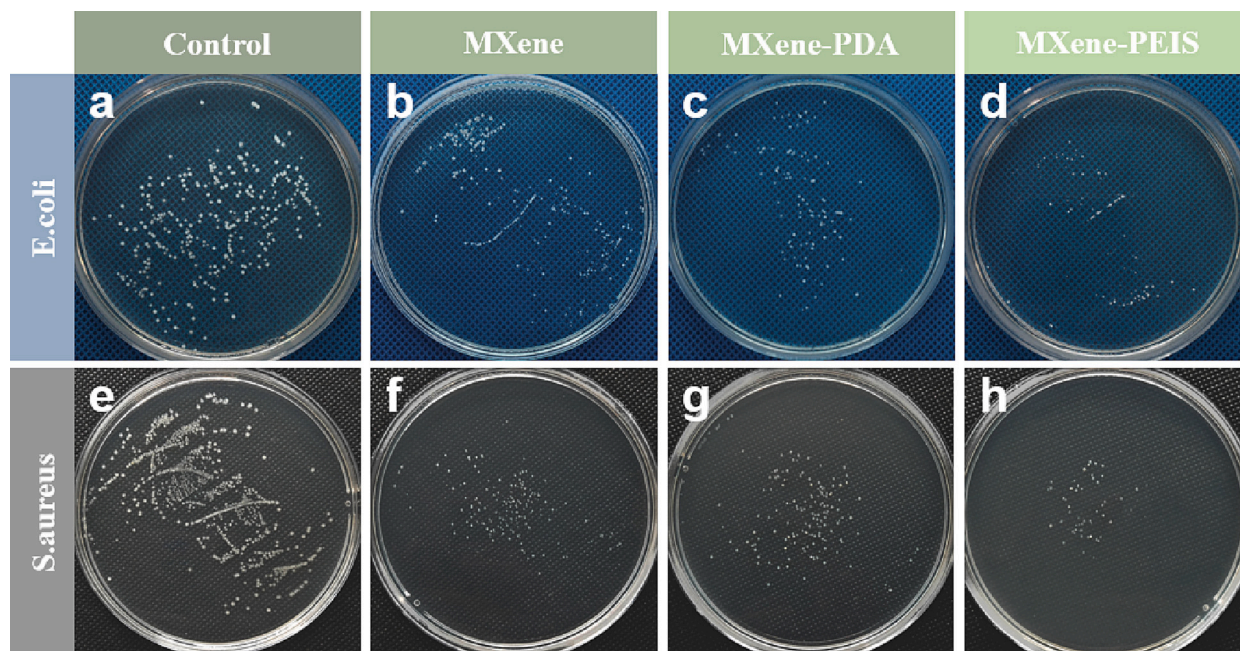


Fig. 3. The growth of *E. coli* and *S. aureus* on the agar surface of each group. (a, e) control; (b, f) MXene; (c, g) MXene-PDA; (d, h) MXene-PEIS.

According to the growing process of marine biofouling, an efficient antifouling strategy will restrain the attachment and growth of fouling species [45]. The antifouling performance of self-polishing nanocomposite coatings was strongly associated with the antifouling effect of the added MXene nanosheets and the hydrolysis of self-polishing coating. The tests for antifouling performance were similar to those for MXene nanosheets (Figs. S4, S5, S6, and S7). Fig. 5 shows the statistical results of antibacterial rate of two typical bacteria (*E. coli* and *S. aureus*) and the adhesion density of two typical microalgae (*Dunaliella* and *Porphyridium*) on the surface of SP, MXene/SP, and MXene-PEIS/SP coatings. As shown in Fig. 5(a), the antibacterial rate of blank self-polishing coatings is about 45 % and 56 % for *E. coli* and *S. aureus*, respectively, indicating a visible anti-bacterial effect. However, the antibacterial rate for MXene/SP increases slightly for *E. coli* but decreases in the case of *S. aureus*. The antibacterial activity of MXene/SP is

unimproved, the probable reason being that the hydrolysis of short-term immersion (5 h) was too slow to release enough MXene nanosheets for the anti-bacterial properties to be effective [46]. Furthermore, the antibacterial rate of MXene-PEIS/SP markedly increased to about 97 % and 84 % respectively for *E. coli* and *S. aureus*, which benefits from both the high hydrophilicity of PEIS accelerating the leaching of MXene-PEIS [47] and the anti-bacterial activity of MXene-PEIS. Therefore, the MXene-PEIS/SP exhibits excellent anti-bacterial performance through the synergistic anti-bacterial effect of the MXene-PEIS and the hydrolysis of self-polishing coatings. As shown in Fig. 5(b), the densities of two typical microalgae (*Dunaliella* and *Porphyridium*) attached on SP, MXene/SP, MXene-PEIS/SP coatings are reducing. When MXene-PEIS is incorporated into the SP coatings, the anti-biofouling performance was significantly promoted, which was facilitated by the synergistic anti-biofouling effect of MXene and the prominent hydration of PEIS as

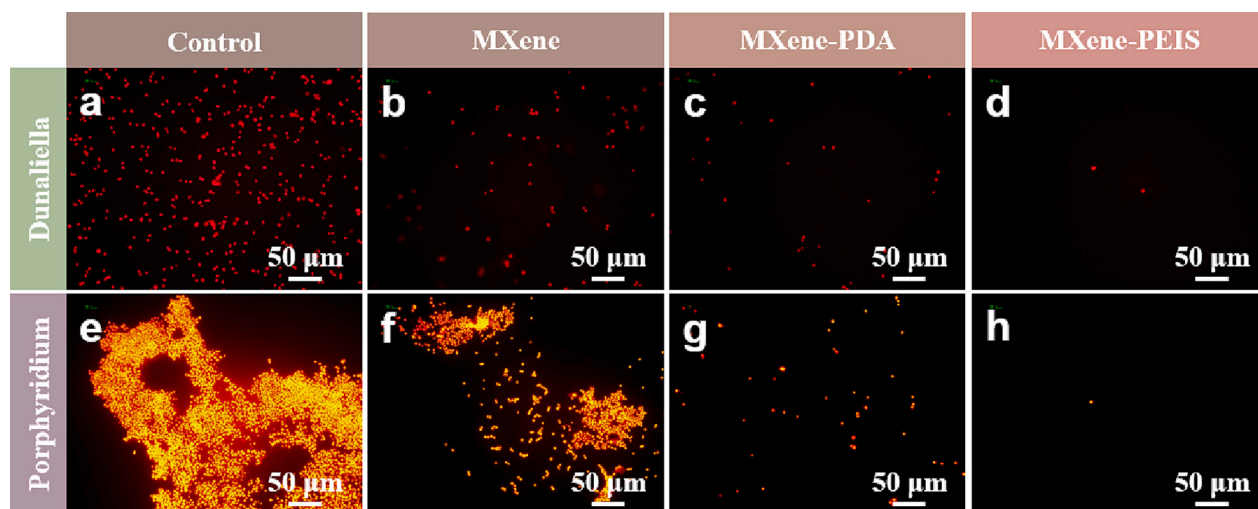


Fig. 4. Fluorescence microscopy images of *Dunaliella* and *Porphyridium* that were adhered on the PDMS film: (a, e) Control; (b, f) MXene; (c, g) MXene-PDA; (d, h) MXene-PEIS.

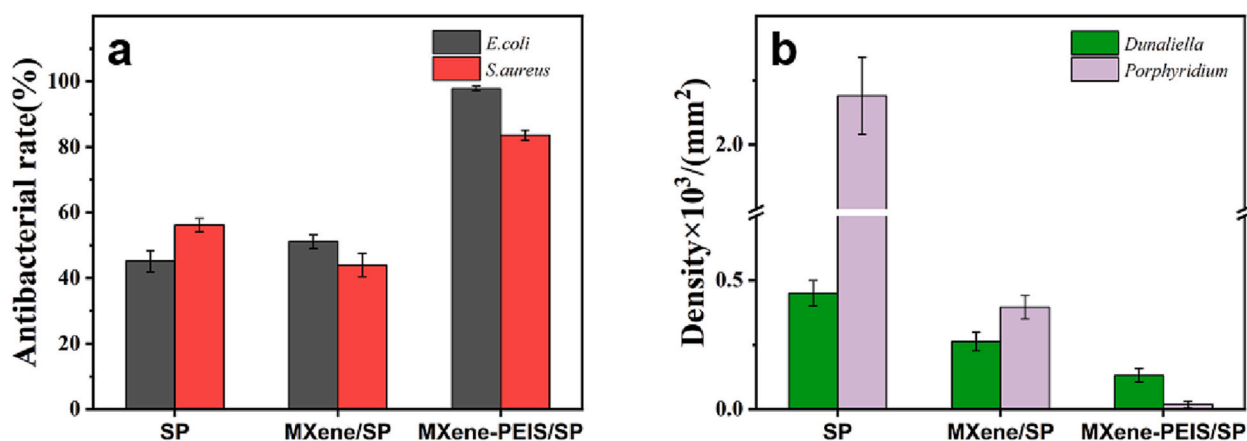


Fig. 5. (a) Antibacterial rate of SP, MXene/SP, and MXene-PEIS/SP for *E. coli* and *S. aureus*; (b) statistical results of *Dunaliella* and *Porphyridium* adhered on the surface of SP, MXene/SP, and MXene-PEIS/SP.

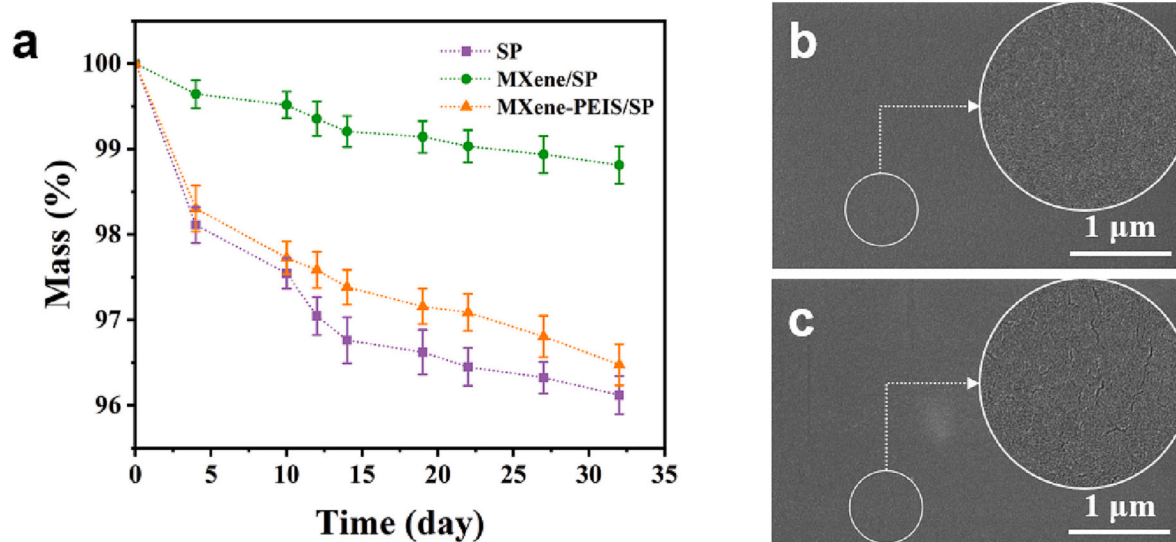


Fig. 6. (a) The hydrolysis over time of SP, MXene/SP, and MXene-PEIS/SP in deionized water; SEM images of MXene-PEIS/SP (b) before and (c) after hydrolysis in deionized water.

well as the hydrolysis of SP coatings. Therefore, MXene-PEIS/SP coatings significantly enhance the antifouling efficiency of common self-polishing coatings.

The self-polishing of SP coatings manifests itself in the decomposition of the coating matrix and the leaching of antifoulant, and relies heavily on the physicochemical properties of antifoulant and coating matrix [48]. The self-polishing rate of SP coatings plays an crucial role in not only the antifouling performance due to the release of antifoulant, but also in prolonging the service life [49]. Fig. 6(a) illustrates the mass change over time of SP, MXene/SP, and MXene-PEIS/SP coatings immersed in deionized water, and Fig. 6(b) and (c) show the SEM images of the surface of MXene-PEIS/SP before and after long-time hydrolysis. After 33-day immersion, the self-polishing rate of MXene/SP was lower than that of SP with a 1.1 % mass loss while for SP the mass loss was 3.9 %. The inhibition of hydrolysis is possibly associated with the combination of the active surface groups of MXene with the coating matrix through covalent bonds and stable adsorption [50], thus affecting the degradation process of the polymer chain in coatings. When PEIS is introduced, the self-polishing rate of MXene-PEIS/SP is obviously higher than that of MXene/SP but slightly lower than that of SP with a 3.4 % mass loss. The contribution to hydrolysis derives from the highly hydrophilic groups of PEIS facilitating the availability of water of dissolution [47]. In addition, the wettability of coating is reported to influence their self-polishing process [48]. As shown in Fig. S10, the water contact angles are no significant differences among the as-prepared coatings before hydrolysis revealing that the incorporation of MXene-PEIS has little effect on surface roughness [51]. After hydrolysis, the MXene-PEIS/SP coatings show the lowest water contact angle (about 57.3°) while that of SP and MXene/SP coatings are respectively about 69.9° and 80.2°, which further proves that MXene-PEIS nanosheets accelerate the hydrolysis via the highly hydrophilic groups of PEIS. Moreover, the differences between the surface of MXene-PEIS/SP coatings before and after hydrolysis are observed in Fig. 6(b) and (c), indicating that self-polishing has happened and many fresh areas remain on the surface for further release of MXene-PEIS nanosheets [45]. Besides, combined with the self-polishing rate of MXene-PEIS/SP, the subtle change in the surface after long-time hydrolysis reveals that the MXene-PEIS nanosheets are beneficial in prolonging the service life of self-polishing coatings. Furthermore, the tensile property of MXene-PEIS/SP coatings slightly degrade after anti-bacterial activity test (Fig. S11), which may be caused by the mild hydrolysis in the surface of coatings. Finally, MXene-PEIS nanosheets proved to have excellent performance in self-polishing coatings with significant improvement in antifouling efficiency and service life.

4. Conclusion

To summarize, we successfully prepared zwitterionic polymer-functionalized MXene nanosheets through a two-step chemical modification method. Firstly, the MXene nanosheets were modified by the in-situ self-assembly of PDA. Then, the PEIS was introduced through Michael addition with PDA. The as-prepared MXene-PEIS exhibits fine anti-bacterial and anti-biofouling performance through the cooperative antifouling effect of MXene nanosheets and zwitterionic polymer PEIS. Besides, the as-prepared MXene-PEIS could be incorporated into the self-polishing coatings to obtain novel nanocomposite antifouling coatings. The results show that MXene-PEIS/SP coatings not only present significant antifouling effects but also have a positive impact on the service life. In conclusion, this approach provides further directions for research into the design of antifouling coatings, as well the further development of antifouling applications of MXene nanosheets.

CRediT authorship contribution statement

Yixin Zeng: Materials synthesis, Characterizations, Formal analysis, Writing-original draft. **Peng Wang:** Characterizations, Data curation,

Formal analysis. **Baoluo He:** Characterizations. **Shujuan Liu:** Review & editing, Project administration, Supervision. **Qian Ye:** Writing-review & editing, Supervision. **Feng Zhou:** Review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2023.107727>.

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